

PART-1 : MATHEMATICS**SECTION-I (i)**

1) Three couples sit for a photograph in 2 rows of three people each such that no couple is sitting the same row next to each other or in the same column one behind the other. How many arrangements are possible ?

- (A) 48
- (B) 56
- (C) 72
- (D) 96

2) Let $\langle U_n \rangle$ and $\langle V_n \rangle$ be two sequences of real numbers such that for every positive integer n , $U_{n+1} = U_n - V_n$ and $V_{n+1} = U_n + V_n$. If $U_{2024} = 2^{1012}$ and $V_{2024} = 2^{1013}$, then $U_1 + 2V_1$ is equal to

- (A) 1
- (B) 3
- (C) 5
- (D) 0

3) The equation $z^{10} + (13z - 1)^{10} = 0$ has 10 complex roots $z_1, \bar{z}_1, z_2, \bar{z}_2, \dots, z_5, \bar{z}_5$. Then the value of $\frac{1}{|z_1|^2} + \frac{1}{|z_2|^2} + \frac{1}{|z_3|^2} + \frac{1}{|z_4|^2} + \frac{1}{|z_5|^2}$ is equal to

- (A) 850
- (B) 1700
- (C) 900
- (D) 1800

4) Let N be the number of all 5 digit numbers each of which contains the block 15 and is divisible by 15. Then the last digit of N is

- (A) 3
- (B) 5
- (C) 7
- (D) 9

SECTION-I (ii)

1) Let $Z = e^{i\left(\frac{2\pi}{19}\right)}$, where $i = \sqrt{-1}$. If $\alpha, \beta, \gamma \in \mathbb{R}$ such that

$$1 + 5Z + 9Z^2 + \dots + 73Z^{18} = \frac{19\alpha}{Z-1}$$

$$1 + 5\cos\left(\frac{2\pi}{19}\right) + 9\cos\left(\frac{4\pi}{19}\right) + \dots + 73\cos\left(\frac{36\pi}{19}\right) = 19\beta$$

$$\& \quad 5\sin\left(\frac{2\pi}{19}\right) + 9\sin\left(\frac{4\pi}{19}\right) + \dots + 73\sin\left(\frac{36\pi}{19}\right) = \gamma \cot\left(\frac{\pi}{19}\right)$$

then which of the following option(s) is/are correct ?

- (A) $(|\beta| + |\gamma|)$ is divisible by 19
 (B) $|\alpha| + |\beta| + |\gamma| = 44$
 (C) $(10\alpha + \beta + \gamma)$ is divisible by 19
 (D) Sum of digits of $(|\alpha| - |\beta| + |\gamma|)$ is 4

2) Which of the following option(s) is/are correct ?

- (A) The number of integers between 10^2 and 10^4 such that the sum of their digits is equal to 14 is 535

- If p, q, r are distinct prime numbers and α, β, γ are positive integers such that L.C.M of α, β, γ is p^3q^2r and greatest common divisor of α, β, γ is pqr , then the number of possible ordered triplets (α, β, γ) is 72.

The sum of all 5-digit numbers formed using the digits 2, 4, 6, 7, 9 without repetition is

- (C) $\frac{(10^5 - 1)(4!)(28)}{9}$

- (D) Suppose there are 6 hats of different colours and 6 shirts of same colour as that of hats. The number of ways in which these can be put on by 5 men such that no man wear the shirt and the hat of the same colour is $310 \times 6!$

3) If $a_n = \sum_{r=0}^n \frac{1}{nCr}$ then $\sum_{r=0}^n \frac{r^2}{nCr} = P(n) a_{n+2} + Q(n) a_{n+1} + a_n + R(n)$ where $P(n), Q(n), R(n)$ are the polynomial functions of n then

- (A) $P(5) = 42$
 (B) $Q(5) = -18$
 (C) $R(5) = -30$
 (D) $P(5) - Q(5) + R(5) = 30$

SECTION-II (i)

Common Content for Question No. 1 to 2

Euler's Formula : $e^{ix} = \cos x + i \sin x$, $x \in \mathbb{R}$. Given that

$$\alpha = \cos \frac{2\pi}{11} + i \sin \frac{2\pi}{11}, \lambda = \alpha^{2019}, \mu = \alpha^{2020}, \beta = \alpha^{2015}$$

Now, answer the following questions :

1) The value of $|i + (i - \beta)(i - \beta^2) \dots (i - \beta^{10})| + 2\operatorname{Re}(\lambda + \lambda^2 + \lambda^3 + \lambda^4 + \lambda^5)$ is equal to

2) The value of $(\alpha - \beta)(\alpha - \beta^2) \dots (\alpha - \beta^{10}) + (\mu - \beta)(\mu - \beta^2) \dots (\mu - \beta^{10})$ is equal to

Common Content for Question No. 3 to 4

Let a curves C_1, C_2 be curves representing locus of point z, w respectively

$$C_1 : |z - 1| = 1 \text{ and } C_2 : w = \frac{z^2 - z - 2}{1 - z}$$

3) Let eccentricity of curve C_2 is $\frac{a\sqrt{2}}{b}$, then the value of $(a + b)$ is

4) Product of slopes of normals to curve C_1 which touch curve C_2 is

SECTION-II (ii)

1) Let $(1 - x^3)^n = \sum_{r=0}^n a_r x^r (1 - x)^{3n-2r} \quad \forall x \in \mathbb{R} - \{1\}$. If $\sum_{n=1}^k \left(\sum_{r=0}^{n-1} {}^k C_n a_r \right) = p^k - q^k$ (where $n, k \in \mathbb{N}$ and p, q are natural number and coprime) then $p + q$ is equal to

2) A person is climbing a staircase containing $3k$ steps ($k \geq 2, k \in \mathbb{N}$). At each move, the person can choose to move up by either 1 step or k steps. Let $A(k)$ denote the total number of distinct ways in which the person can reach the top step of the staircase, starting from the bottom of the staircase. If a sequence $\langle \lambda_k \rangle$ is defined as $\lambda_k = A(k + 1) - A(k)$ for all $k \geq 2$ then the value of $\lambda_{k+1} - \lambda_k$ is equal to

3) The remainder when $\sum_{k=0}^{1012} \left(\frac{\binom{1012}{k}}{\binom{2024}{k}} \sum_{r=k}^{2024} \left(\binom{r}{k} \binom{2024}{r} \right) \right)$ is divided by 49 is
 (where $\binom{n}{r}$ denotes ${}^n C_r$)

4) Let z_1, z_2, z_3 be complex numbers of equal magnitude satisfying $z_1 + z_2 + z_3 = -\frac{\sqrt{3}}{2} - i\sqrt{5}$ and $z_1 z_2 z_3 = \sqrt{3} + i\sqrt{5}$. If $z_1 = x_1 + iy_1, z_2 = x_2 + iy_2, z_3 = x_3 + iy_3$; $x_1, x_2, x_3, y_1, y_2, y_3 \in \mathbb{R}$, where $i = \sqrt{-1}$, then the value of $\frac{16}{5} (x_1 y_1 + x_2 y_2 + x_3 y_3)^2$ is

5) Let $A = \sum_{r=1}^{50} \frac{{}^{50+r} C_r (2r-1)}{{}^{50} C_r (50+r)}$, $B = \sum_{r=0}^{50} ({}^{50} C_r)^2$ and $C = \sum_{r=1}^{49} \frac{2r^2 - 48r + 1}{(50-r) {}^{50} C_r}$ then value of $A - B + C$ is

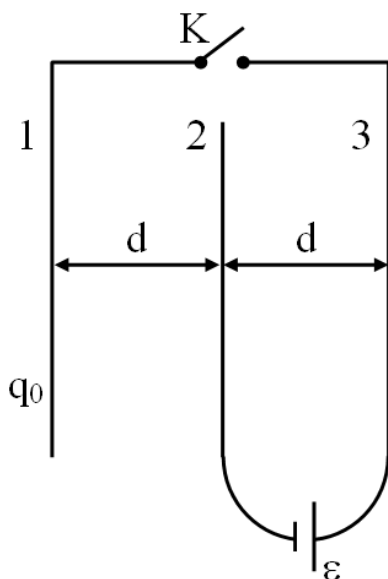
6) The polynomial $P(x) = (1 + x + x^2 + \dots + x^{17})^2 - x^{17}$ has 34 distinct complex roots of the form $Z_k = r_k e^{i(2\pi a_k)}$, where $k = 1, 2, 3, 4, \dots, 34$ with $0 < a_1 < a_2 < a_3 < \dots < a_{34} < 1$ and $r_k > 0$. Given that $a_1 +$

$a_2 + a_3 + a_4 + a_5 = \frac{m}{n}$ where m and n are relatively prime natural numbers then value of $m + n$ is

PART-2 : PHYSICS

SECTION-I (i)

1) Three identical thin metallic plates 1, 2 and 3, each of area S , are arranged parallel to one another. The separation between adjacent plates is d , where d is much smaller than the linear dimensions of the plates. Initially, all three plates are uncharged. A battery of emf ε is connected between plates 2 and 3, with its positive terminal connected to plate 3. Plate 1 is then given a charge q_0 . The switch K , which connects plates 1 and 3, is closed. Neglecting edge effects, the final charge



on plate 3 is

- (A) $\frac{q_0}{2} - \frac{\varepsilon_0 S \varepsilon}{d}$
- (B) $q_0 + \frac{\varepsilon_0 S \varepsilon}{d}$
- (C) $\frac{q_0}{2} + \frac{\varepsilon_0 S \varepsilon}{d}$
- (D) $\frac{q_0}{2} + \frac{2\varepsilon_0 S \varepsilon}{d}$

2) A metallic resistor has resistance 10Ω at 20°C and temperature coefficient of resistance $\alpha = 5.0 \times 10^{-3}\text{K}^{-1}$. It is connected to a 60V battery having internal resistance 10Ω . Due to Joule heating, the temperature of the resistor rises above the surroundings, which remain at 20°C . The rate of heat loss from the resistor to the surroundings is $P_{\text{loss}} = 0.40(T - 20)\text{W}$, where T is the resistor temperature in $^\circ\text{C}$. Assume $R(T) = R_{20}[1 + \alpha(T - 20)]$ and neglect thermal expansion. For $x = 1 + 0.005(T - 20)$, the equilibrium condition leads to $2x^3 + 2x^2 - 11x - 2 = 0$. The roots of this equation

are $x = 2$, $x = \frac{-3 + \sqrt{7}}{2}$, $x = \frac{-3 - \sqrt{7}}{2}$. At thermal equilibrium, the temperature of the resistor is

- (A) 120°C
- (B) 170°C

- (C) 220°C
 (D) 270°C

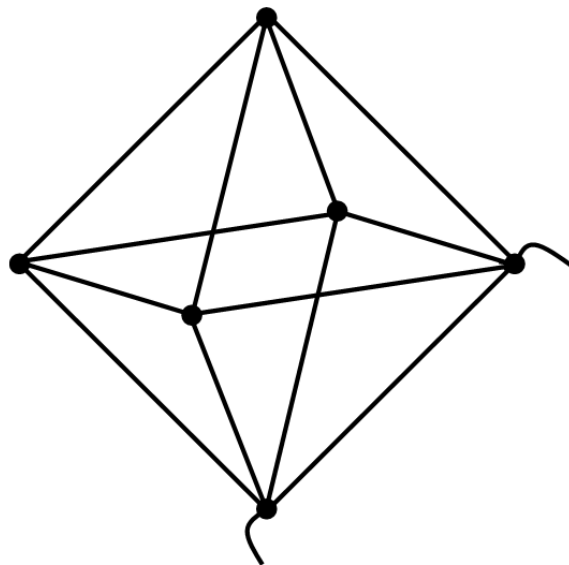
3) A moving-coil galvanometer has resistance $G_0 = 100\ \Omega$ at 20°C and requires a current $I_g = 2.0\ \text{mA}$ for full-scale deflection. The galvanometer coil has temperature coefficient $\alpha_g = +4.0 \times 10^{-3}\text{K}^{-1}$. It is to be converted into a $0 - 13\ \text{V}$ voltmeter by connecting two resistors R_1 and R_2 in series with it. At 20°C , their temperature coefficients are $\alpha_1 = +2.0 \times 10^{-3}\text{K}^{-1}$, $\alpha_2 = -1.0 \times 10^{-3}\text{K}^{-1}$. All resistances may be assumed to vary linearly with temperature. The full-scale current I_g is temperature independent. The voltmeter is required to satisfy both conditions :

1. Its full-scale range is $13\ \text{V}$ at 20°C .
2. Its full-scale voltage is independent of temperature to first order about 20°C .

The values of R_1 and R_2 at 20°C are :

- (A) $R_1 = 2.0\ \text{k}\Omega$, $R_2 = 4.4\ \text{k}\Omega$
 (B) $R_1 = 2.4\ \text{k}\Omega$, $R_2 = 4.0\ \text{k}\Omega$
 (C) $R_1 = 3.2\ \text{k}\Omega$, $R_2 = 3.2\ \text{k}\Omega$
 (D) $R_1 = 4.4\ \text{k}\Omega$, $R_2 = 2.0\ \text{k}\Omega$

4) An octahedron is constructed by soldering together twelve identical wire segments, each having resistance R . Find the equivalent resistance of the octahedral network when an ohmmeter is



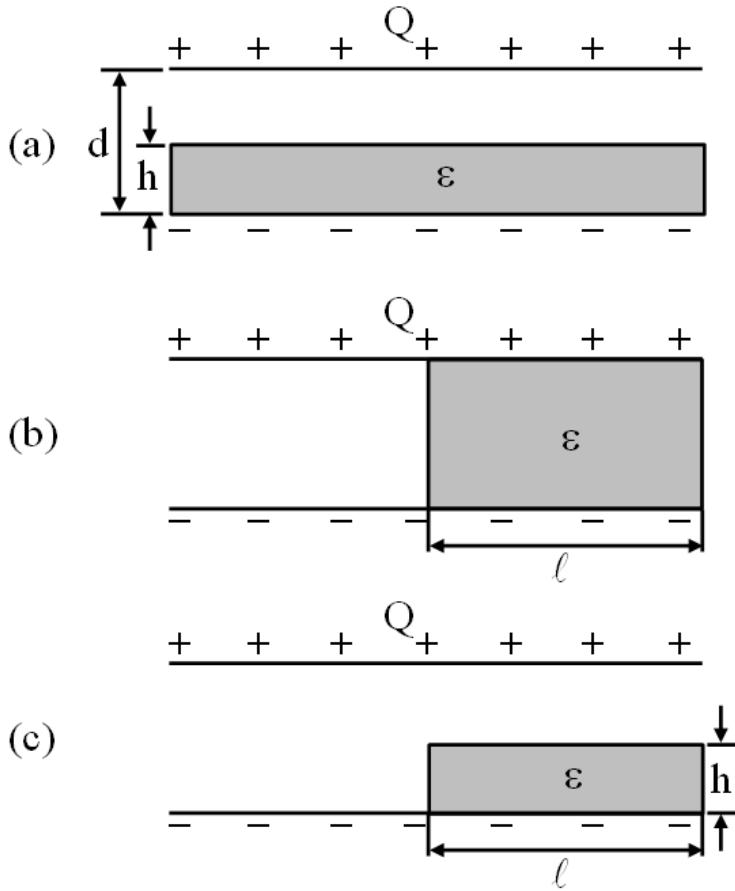
connected to the two terminals shown in the figure.

- (A) $\frac{5}{12}R$
 (B) $\frac{12}{5}R$
 (C) $\frac{10}{19}R$
 (D) $\frac{19}{10}R$

SECTION-I (ii)

1) An isolated parallel-plate capacitor has square plates of area S and separation d . The plates carry

fixed charges $+Q$ and $-Q$. A dielectric of permittivity ϵ is inserted in three different configurations as shown in the figure. The permittivity of air is ϵ_0 . Neglect fringing effects. In cases (a) and (c), the dielectric has thickness h . In case (b) and (c), its horizontal extent is ℓ . Since each plate is square, its side is $\sqrt{S}$. Let E_a , E_b , E_c be the magnitudes of the electric field inside the dielectric in cases (a), (b) and (c), respectively. Which of the following statements are correct ?



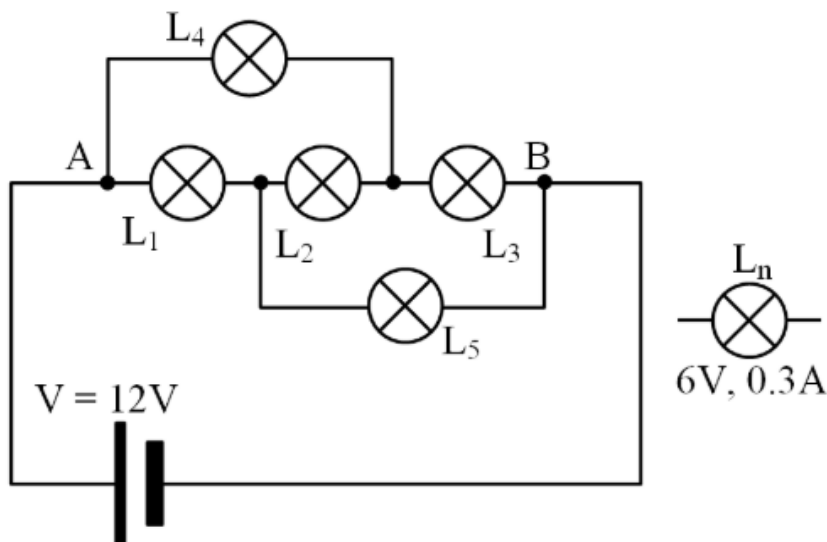
(A) In case (a), the normal component of electric field remains the same in the air and dielectric layers. Hence, $E_a = \frac{Q}{\epsilon S}$.

(B) In case (b), the air-filled and dielectric-filled portions are effectively connected in parallel and experience the same potential difference. Therefore, $E_b = \frac{Q}{\epsilon \ell \sqrt{S} + \epsilon_0 (S - \ell \sqrt{S})}$.

(C) In case (c), the portion of area $\ell \sqrt{S}$ containing the dielectric behaves as a series combination of air and dielectric layers, while the remaining portion is purely air-filled. Consequently, $E_c = \frac{Q}{\epsilon \ell \sqrt{S} + (S - \ell \sqrt{S}) \frac{\epsilon(d-h) + \epsilon_0 h}{d}}$.

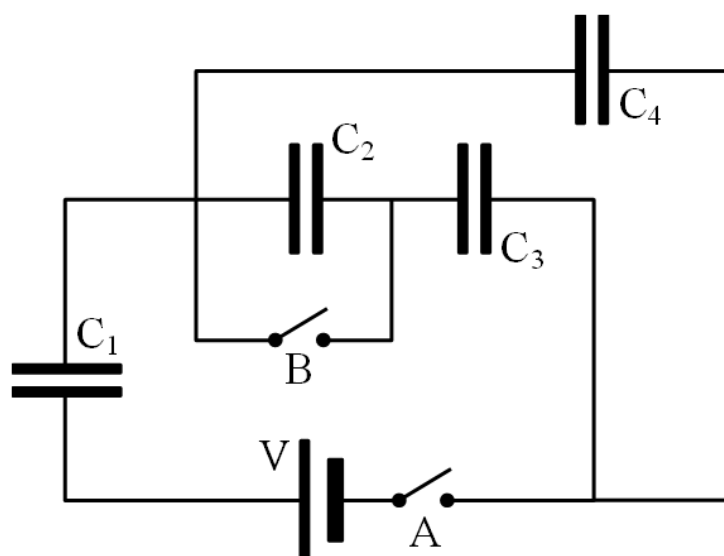
(D) In case (c), increasing the dielectric thickness h , while keeping Q , S , d , ℓ , and ϵ fixed, does not alter the electric field inside the dielectric.

2) Five identical incandescent bulbs L_1 , L_2 , L_3 , L_4 , L_5 are connected to an ideal 12 V battery as shown in the figure. Each bulb is rated 6 V, 0.3 A. For circuit calculations, each intact bulb may be treated as an ohmic resistor equal to its rated resistance. A bulb remains dark at 0V, glows dimly for $0 < V < 6$ V, glows normally at 6 V, and glows brighter than normal for $6 < V \leq 8.5$ V. It burns out if the voltage across it exceeds 8.5 V. Whenever a bulb is said to be removed, its branch is to be treated as open-circuited. Which of the following statements are correct ?



- (A) With all five bulbs connected, L_2 remains dark, while L_1 , L_3 , L_4 and L_5 each glow with normal brightness.
- (B) If L_2 is removed, the currents and brightnesses of all the other bulbs remain unchanged.
- (C) If L_1 alone is removed, the voltages across L_2 , L_3 , L_4 and L_5 are respectively 2.4 V, 4.8 V, 7.2V and 2.4 V. Thus, L_4 becomes brighter than normal while the other three glow dimly.
- (D) If L_4 alone is removed, the voltage across L_1 exceeds 8.5 V, so L_1 immediately burns out.

3) In the capacitor network shown, all capacitors are initially uncharged. Switch A is closed while B remains open. After steady state is reached, B is also closed while A remains closed. Given $V = 15$ V, $C_1 = C = 3\mu\text{F}$, $C_2 = 2C$, $C_3 = 4C$, $C_4 = 2C$. Which of the following statements are correct ?



- (A) When B is open, the network behaves as an equivalent capacitor of $C_{\text{eq}} = \frac{30}{13}\mu\text{F}$, so the total electrostatic energy stored in the network is $U = \frac{3375}{13}\mu\text{J}$.
- When B is open, conservation of charge at the isolated internal junctions given the charges
- (B) $(Q_1, Q_2, Q_3, Q_4) = \left(\frac{450}{13}, \frac{180}{13}, \frac{180}{13}, \frac{270}{13}\right)\mu\text{C}$.

- Closing B short-circuits C_2 , so its final potential difference becomes zero. After charge redistribution, the total electrostatic energy of the capacitor network increases by $\frac{2700}{91} \mu\text{J}$.
- (C) Since the battery remains connected while B is closed, charge is exchanged with the source
- (D) during the redistribution. The net additional charge supplied by the battery is $\frac{360}{91} \mu\text{C}$.

SECTION-II (i)

Common Content for Question No. 1 to 2

A nonlinear resistive element exhibits thermal hysteresis. Its resistance depends on temperature as follows:

- While heating, the resistance remains $R_1 = 50\Omega$ until the temperature reaches 100°C , at which point it changes instantaneously to $R_2 = 100\Omega$.
- While cooling, the resistance remains $R_2 = 100\Omega$ until the temperature falls to 99°C , at which point it changes instantaneously back to $R_1 = 50\Omega$.

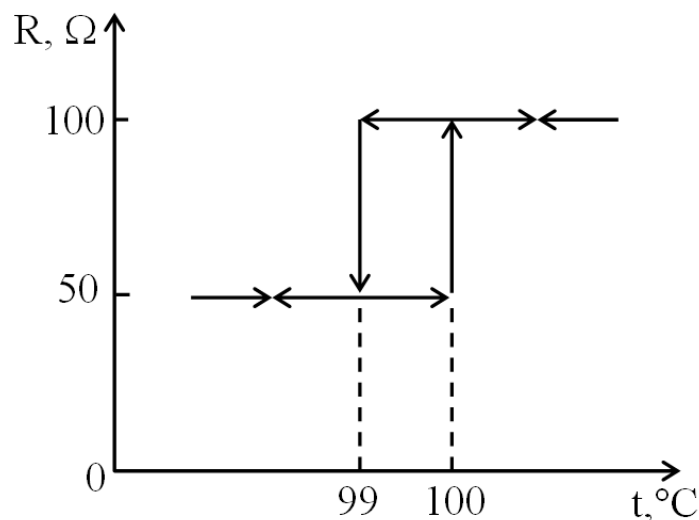
The ambient temperature is 20°C . The rate of heat loss to the surroundings is proportional to the temperature difference:

$$P_{\text{loss}} = k(T - 20),$$

where k is a constant.

When a constant voltage of 60 V is applied, the element reaches a steady temperature of 80°C . The heat capacity of the element is $C = 3\text{ J K}^{-1}$.

The voltage is then changed to 80 V . The element repeatedly heats from 99°C to 100°C in the 50Ω state and cools from 100°C to 99°C in the 100Ω state, producing self-sustained current oscillations. Assume that the element is always at a uniform temperature. Take $\ln(1 + x) \approx x$



- 1) Determine the period of the self-sustained oscillations, in seconds. Give your answer to two decimal places.
- 2) Determine the maximum current during the oscillations, in amperes.

Common Content for Question No. 3 to 4

A rectangular glass plate of thickness $h = 0.50\text{ mm}$ has metallic coatings on its two faces. The lower face is completely coated, whereas the upper face is coated only over part of its area. The two coatings form a capacitor with glass as dielectric.

A small drop of a conducting liquid is placed on the uncoated portion of the upper surface. The liquid is completely non-wetting (contact angle 180°) with respect to glass. The upper metallic coating and the conducting liquid are electrically connected, so they remain at the same potential.

This capacitor is connected in series with a reference capacitor of fixed capacitance C_0 , and the combination is connected to a variable DC source of voltage U . The voltage U_c across the reference capacitor is measured while U is gradually increased from zero.

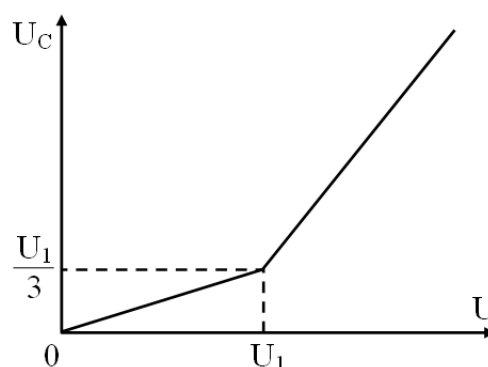
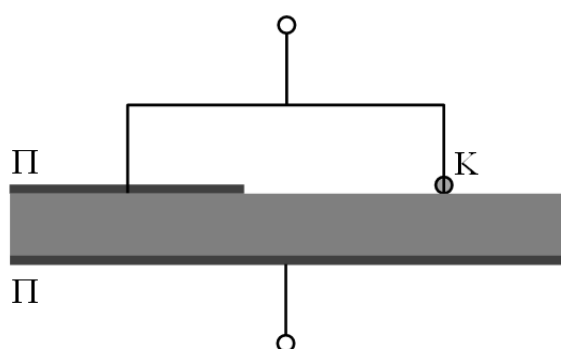
The experimental U_c -versus- U graph consists of two straight-line segments. The change in slope

occurs at $U = U_1$, and at this instant, $U_c = \frac{U_1}{3}$.

Beyond this point, the electric field causes the conducting liquid to spread over the glass. During this spreading, mechanical equilibrium between electrostatic and surface-energy effects maintains the voltage across glass capacitor at its threshold value.

For the liquid, the surface-tension coefficients at the liquid-air and liquid-glass interfaces are equal to σ . Neglect gravity and fringing effects.

Take : $U_1 = 5.90$ kV, $\epsilon_r = 7$, $\epsilon_0 = 8.85 \times 10^{-12}$ F m $^{-1}$.



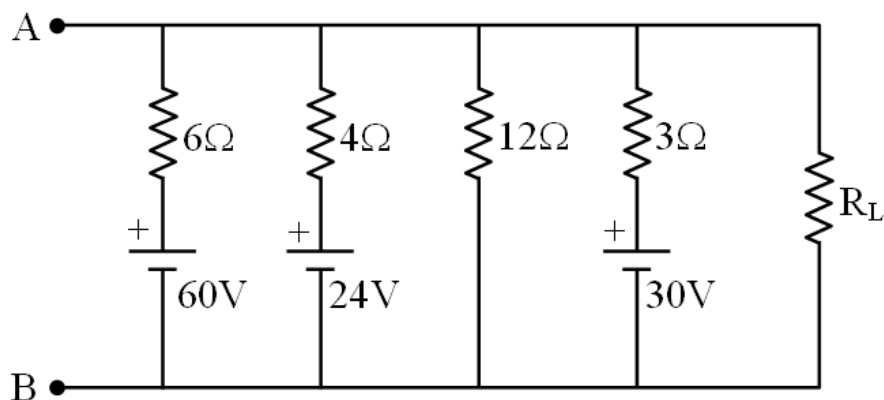
3) When the source voltage is increased to $U = 2U_1$, find the voltage U_c across the reference capacitor. Give your answer in kV, rounded to two decimal places.

4) Using the threshold condition for spreading of the conducting liquid, determine its surface-tension coefficient σ . Give your answer in N m $^{-1}$, rounded to two decimal places.

SECTION-II (ii)

1) Three batteries of emfs $\epsilon_1 = 6$ V, $\epsilon_2 = 5$ V, $\epsilon_3 = 3$ V and internal resistances $R_1 = 2 \Omega$, $R_2 = 1 \Omega$, $R_3 = 4 \Omega$ are connected in a closed ring A-B-C-A. The batteries are oriented so that their emfs aid one another in the clockwise direction $A \rightarrow B \rightarrow C \rightarrow A$. Three ideal capacitors $C_1 = 1 \mu\text{F}$, $C_2 = 5 \mu\text{F}$, $C_3 = 6 \mu\text{F}$ have their outer plates connected to A, B, C, respectively, while their inner plates are connected to an isolated common point O. After steady state is reached, a charge $Q = 48 \mu\text{C}$ is deposited on O. Neglect leakage. Find the magnitude of the final charge, in μC , on capacitor C_3 .

2) A variable load resistance R_L is connected across A and B. Determine the maximum power, in watts, that can be delivered to R_L . Enter the nearest integer.



3) A metallic conductor of length $L = 1.0 \text{ m}$ has a non-uniform cross-sectional area given by $A(x) = A_0 \left(1 + \frac{x}{L}\right)^2$, where x is measured from one end and $A^0 = 1.0 \text{ mm}^2$. The metal is monovalent, has density $8.0 \times 10^3 \text{ kg m}^{-3}$, and molar mass 64 g mol^{-1} . A steady current of 3.84 A flows through the conductor when a potential difference of 0.20 V is applied between its ends. Assume that the electron mobility μ is independent of position and that each atom contributes one conduction electron. Given : $N_A = 6.0 \times 10^{23} \text{ mol}^{-1}$, $e = 1.6 \times 10^{-19} \text{ C}$, find the value of μ in $\text{cm}^2 \text{ V}^{-1} \text{ s}^{-1}$. Enter nearest integer.

4) A meter bridge has a 100 cm long wire of uniform resistivity, but its cross-sectional area varies

$$A(x) = \frac{A_0}{1 + \frac{\alpha x}{L}}, \quad L = 100 \text{ cm}, \text{ where } A^0 \text{ and the positive constant } \alpha \text{ are unknown.}$$

An unknown resistance X is placed in the left gap and a 10Ω resistance in the right gap. The null point is obtained at $\ell_1 = 50.0 \text{ cm}$ from the left end. The positions of X and the 10Ω resistance are then interchanged, without reversing the bridge wire. The new balance point is $\ell_2 = 72.47 \text{ cm}$ from the same left end. Neglect the resistance of connecting leads, end corrections and contact resistances. The jockey draws negligible current at balance. Determine the value of X , in ohms.

5) A galvanometer of resistance 99Ω gives full-scale deflection at a current of 1 mA . It is converted into a 100 mA ammeter by connecting a uniform resistance wire as a shunt across it. The same shunt wire is now cut into three equal parts, and all three parts are connected in parallel with each other. This parallel combination is then connected as the new shunt across the galvanometer. Neglect the resistance of connecting wires. Determine the new full-scale range of the ammeter, in mA .

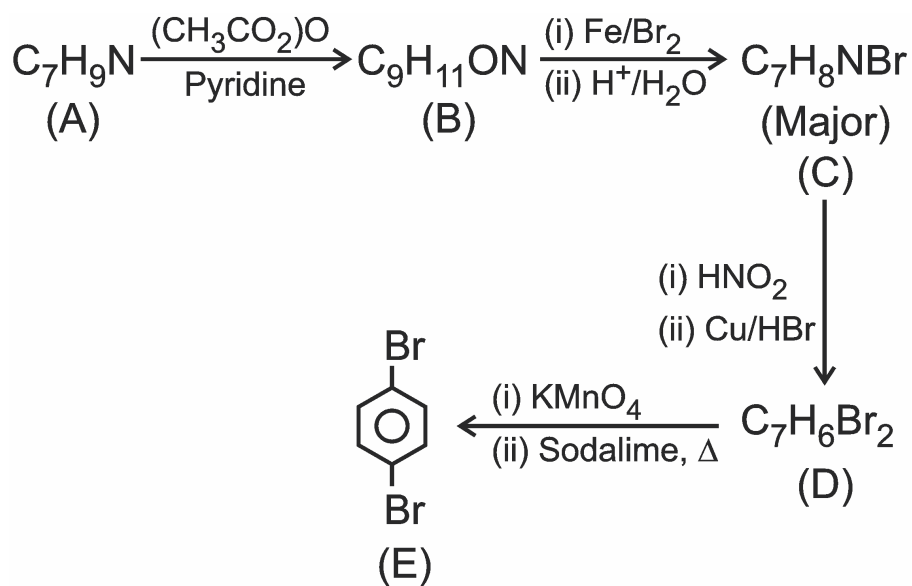
6) A capacitor network is connected between terminals A and B. Two internal junctions are P and Q. The capacitors are connected as follows :

$$C_{AP} = 2 \mu\text{F}, \quad C_{PB} = 5 \mu\text{F},$$

$$C_{AQ} = 3 \mu\text{F}, \quad C_{QB} = 3 \mu\text{F},$$

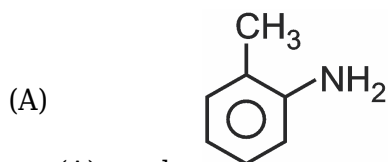
and a capacitor of capacitance $C_{PQ} = 3 \mu\text{F}$ is connected between P and Q. The network is initially uncharged. A potential difference is applied between A and B. Determine the equivalent capacitance of the network between A and B, in μF .

SECTION-I (i)

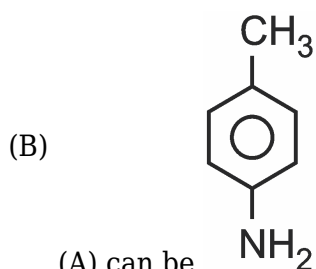


1)

(A) gives positive carbylamine test. Choose the correct option.



(A) can be

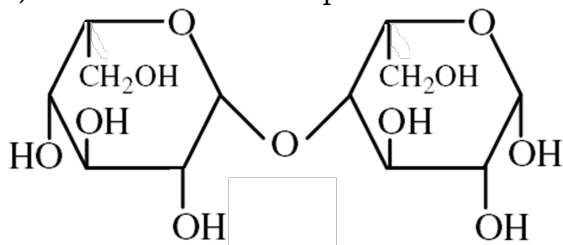


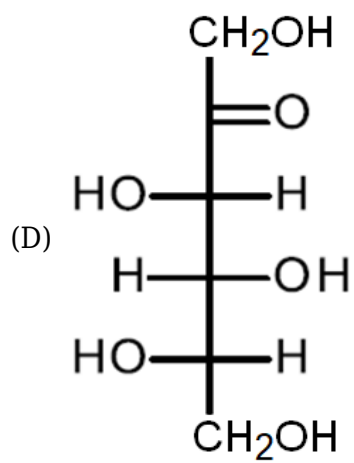
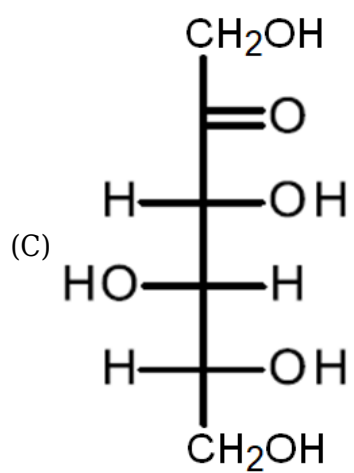
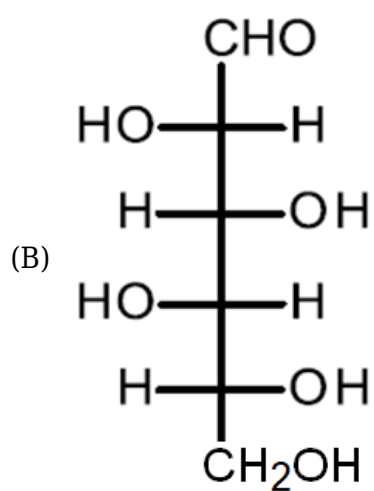
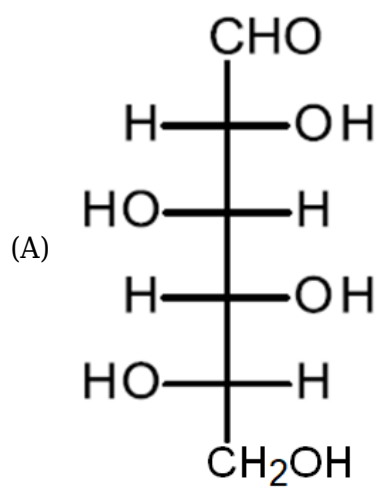
(A) can be

(C) The above conversion involves Sandmeyer reaction

(D) (C) is more basic than aniline

2) The monosaccharide present in the following disaccharide is





3) A colourless substance 'P' (C_6H_7N) is sparingly soluble in water, but when treated with mineral acid, gives 'Q' which is water soluble. 'P' on heating with $CHCl_3 + KOH$, produces a foul smelling compound 'R'. 'P' also reacts with $PhSO_2Cl$ and produces compound 'S' which is soluble in alkali. 'P' reacts with HNO_2 at $0^\circ C$ to produce 'T', which on reaction with β -naphthol in alkaline medium gives a red-orange dye. Select the correct statement(s) about (P)

- (A) (P) on acylation with acetic anhydride gives benzanilide
- (B) (P) cannot give Friedel Crafts acylation
- (C) (P) can be obtained by Gabriel phthalimide synthesis reaction using appropriate reagents
- (D) (P) reacts with (T) in alkaline medium to give yellow dye

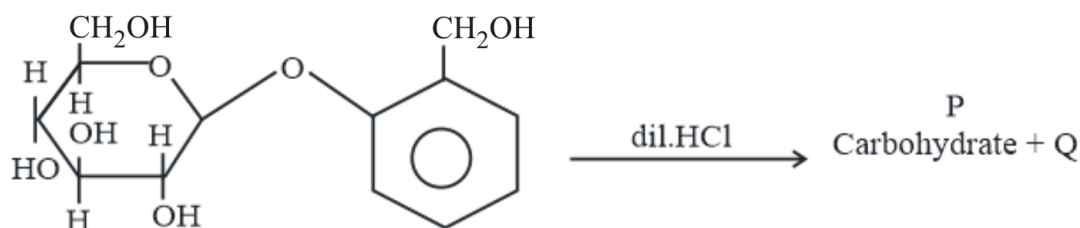
4) Select the correct statement-

- (A) Glucose & Fructose can be differentiated by Barfoed's test
- (B) Ribose & Mannose can be differentiated by Seliwanoff's test
- (C) Valine & Proline can be distinguished by Xanthoproteic test
- (D) Alanine & Insulin can be differentiated by Biuret test

SECTION-I (ii)

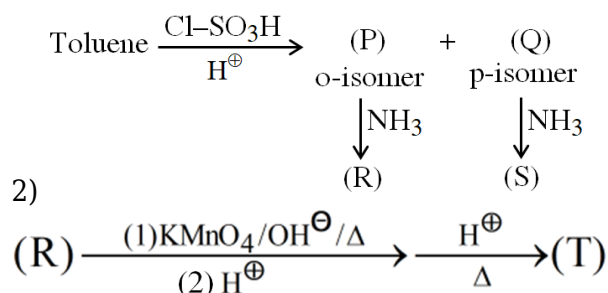
1) Salicin (Structure given below) is a glycoside, found in the bark of willow tree, used in relieving pain.

Observe the following reaction of salicin



The correct statement (s) is (are)

- (A) P is D - glucose
- (B) Q is a non-narcotic analgesic
- (C) Q can be converted to aspirin by side chain oxidation, followed by acetylation.
- (D) The above reaction occurs through a carbocation intermediate



Select the correct statement(s)-

- (A) (Q) can be used to convert -OH to good leaving group
- (B) (S) contains the functional group sulfonamide, which forms the basis of several drugs
- (C) (T) can be used as tranquilizer
- (D) The degree of unsaturation of (T) is 7

3)

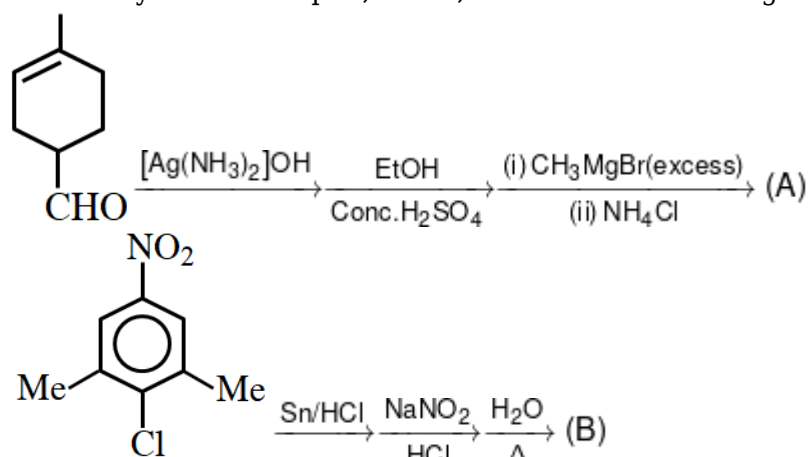
Select the correct statement(s)-

- (A) Vulcanisation of rubber increases cross-links and makes the rubber stiffened
- (B) Low density polythene is formed by polymerisation of ethene using Zeigler-Natta catalyst at 333K-343K and 6-7 atm
- (C) PHBV is a biodegradable polymer
- (D) Novolac is a linear polymer used in paints

SECTION-II (i)

Common Content for Question No. 1 to 2

Commonly used antiseptic, dettol, is a mixture of two organic compounds (A) & (B).



- 1) For (A), calculate the value of $x + y$, where
 x = total stereoisomers (If it has no stereoisomer, report $x = 0$)
 y = total number of carbon atoms present in parent chain, if it is named as per IUPAC rules.

- 2) If (B) is named as per IUPAC rules, calculate sum of locants of all substituents

Common Content for Question No. 3 to 4

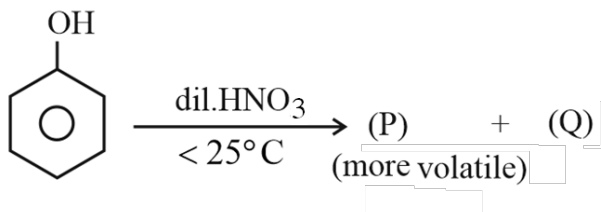
Aspirin on acidic hydrolysis gives two organic compounds (P) & (Q). (P) gives positive FeCl_3 test. (Q) on reaction with SOCl_2 gives (R). (R) on reaction with diazomethane, followed by heating with Ag_2O and hydrolysis gives (S). (S) on reaction with AgOH followed by Br_2/CCl_4 forms (T). (T) on reaction with AgCN gives (U), which on acidic hydrolysis gives an acid (V) and a salt (W). The salt (W) on basic workup gives (X), which gives mustard oil smell with $\text{CS}_2/\text{HgCl}_2$. (X) on reaction with nitrous acid, followed by hydrolysis gives (Y).

Given, atomic mass : $\text{H} = 1$, $\text{C} = 12$, $\text{N} = 14$, $\text{O} = 16$, $\text{S} = 32$, $\text{Cl} = 35.5$, $\text{Br} = 80$

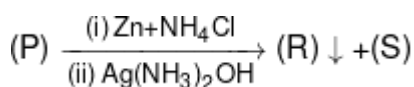
- 3) Calculate the molecular mass of product obtained on reaction of (P) with bromine water

4) (Y) on oxidation with Jones reagent, followed by heating with ammonia gives (Z). (Z) on reaction with Br_2/NaOH followed by acylation with benzoyl chloride gives an aromatic compound (A). Calculate the molecular mass of (A)

SECTION-II (ii)

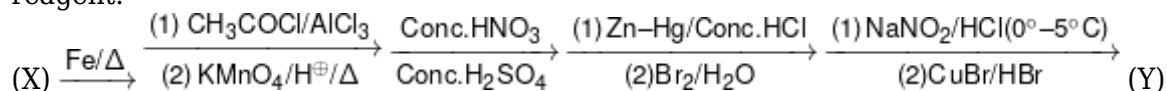


1)
(P, Q are isomers)



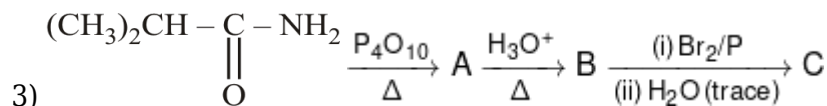
How many oxygen atoms are present in one molecule of the organic product (S)

2) (X) is the smallest hydrocarbon (composed of C^{12} & H^1 only) that can form precipitate with Tollen's reagent.



Starting with 1 mole (X), calculate the mass of major organic product (Y) produced (in grams), assuming 100% yield at each step.

[Given, atomic mass : H = 1, C = 12, N = 14, O = 16, S = 32, Cl = 35, Br = 80]



Calculate the molecular mass of the organic product (C)

[H = 1, C = 12, N = 14, O = 16, P = 31, Br = 80]

4)

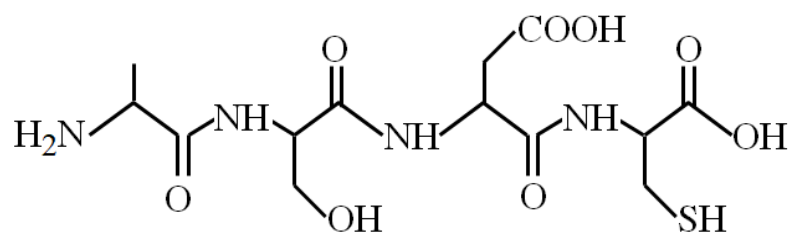
Consider the following polymers

- | | |
|-----------------------|----------------------|
| (1) Polythene | (2) Nylon-6,6 |
| (3) Buna-N | (4) Buna-S |
| (5) Neoprene | (6) PVC |
| (7) Bakelite | (8) Teflon |
| (9) Polyacrylonitrile | (10) Terylene |
| (11) Novalac | (12) Nylon-2-nylon-6 |

How many of the above are homo polymers ?

5) A nucleoside is formed using ribose sugar and uracil base. Calculate the total number of nitrogen and oxygen atoms present in the nucleoside.

6) How many essential amino acids are obtained on hydrolysis of the following tetrapeptide



ANSWER KEYS

PART-1 : MATHEMATICS

SECTION-I (i)

Q.	1	2	3	4
A.	D	C	A	D

SECTION-I (ii)

Q.	5	6	7
A.	B,C,D	A,B	A,B,C,D

SECTION-II (i)

Q.	8	9	10	11
A.	1.00	0.00	5.00	-3.00

SECTION-II (ii)

Q.	12	13	14	15	16	17
A.	9	1	22	3	2498	482

PART-2 : PHYSICS

SECTION-I (i)

Q.	18	19	20	21
A.	C	C	A	A

SECTION-I (ii)

Q.	22	23	24
A.	B,C	A,B,C	A,B,C,D

SECTION-II (i)

Q.	25	26	27	28
A.	0.18 to 0.20	1.60	7.87	0.48

SECTION-II (ii)

Q.	29	30	31	32	33	34
A.	14	203	8	6	892	3

PART-3 : CHEMISTRY

SECTION-I (i)

Q.	35	36	37	38
A.	A	A	B	D

SECTION-I (ii)

Q.	39	40	41
A.	A,C,D	A,B	A,C,D

SECTION-II (i)

Q.	42	43	44	45
A.	5.00	12.00	331.00	135.00

SECTION-II (ii)

Q.	46	47	48	49	50	51
A.	2	146	167	5	8	0

SOLUTIONS

PART-1 : MATHEMATICS

1) Case-I	Row 1	all 3 husbands and wife de arranged in Row 2
		$3! \times 2 = 12$
Case-II	Row 1	all 3 wife and husband de-arranged in Row 2
		$3! \times 2 = 12$
Case-III	Row 1	two husband and other wife and rest de-arranged in Row 2
		${}^3C_2 \times 3! \times 2 = 3 \times 6 \times 2 = 36$
Case-IV	Row 1	two wife and other husband and rest de-arranged in Row 2
		${}^3C_2 \times 3! \times 2 = 3 \times 6 \times 2 = 36$
Total		$36 + 36 + 12 + 12 = 96$

2) Consider $U_{n+1} + i V_{n+1} = (1 + i)(U_n + i V_n)$

Put $n = 1$ $U_2 + i V_2 = (1 + i)(U_1 + i V_1)$

$n = 2$ $U_3 + i V_3 = (1 + i)(U_2 + i V_2)$

$\vdots$

$n = 2023$ $U_{2024} + i V_{2024} = (1 + i)(U_{2023} + i V_{2023})$

Multiply $U_{2024} + i V_{2024} = (1 + i)^{2023} (U_1 + i V_1)$ (1)

$$(1 + i)^{2023} = \left(\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \right)^{2023}$$

$$= 2^{1011} (1 - i)$$

Put in equation (1)

$$2^{2012} + i 2^{2013} = 2^{1011} (1 - i) (U_1 + i V_1)$$

$$-1 + 3i = U_1 + i V_1$$

$$\Rightarrow U_1 + 2V_1 = 5$$

$$3) \left(13 - \frac{1}{z} \right)^{10} + 1 = 0 \Rightarrow 13 - \frac{1}{z} = e^{i \left(\frac{\pi + 2k\pi}{10} \right)} \quad k = 0, 1, 2, \dots, 9$$

$$\Rightarrow \frac{1}{z} = 13 - \cos \frac{(2k+1)\pi}{10} - i \sin \frac{(2k+1)\pi}{10}$$

$$\frac{1}{z \bar{z}}$$

$$\frac{1}{z \bar{z}} = (13 - \cos \theta)^2 + \sin^2 \theta = 170 - 26 \cos \theta$$

$$\sum_{r=1}^5 \frac{1}{|z_r|^2} = 170 \times 5 - 26 \left(\cos \frac{\pi}{10} + \cos \frac{3\pi}{10} + \cos \frac{5\pi}{10} + \cos \frac{7\pi}{10} + \cos \frac{9\pi}{10} \right)$$

$$= 850 - 26(0) = 850$$

4) Various forms of the numbers can be

(i) abc15 (ii) ab150 (iii) ab155 (iv) a15b0

(v) a15b5 (vi) 15ab0 (vii) 15ab5

Number of numbers of form abc15 = 300 {102, 105, ..., 999}

Number of numbers of form ab150 = 30 {12, 15, ..., 99}

Number of numbers of the form ab155 = 30 {10, 13, ..., 97}

Number of numbers of the form a15b0 = 30 {12, 15, ..., 99}

Number of numbers of the form $a15b5 = 30 \{10, 13, \dots, 97\}$
 Number of numbers of the form $15ab0 = 34 \{00, 03, \dots, 99\}$
 Number of numbers of the form $15ab5 = 33 \{01, 04, \dots, 97\}$
 Number of common numbers in number of form $abc15$ & $a15b5 = 3 \{31515, 61515, 91515\}$
 Number of common numbers in number of form $ab150$ & $15ab0 = 1 \{15150\}$
 Number of common numbers in number of form $ab155$ & $15ab5 = 1$ (Reject) (Not div. by 3)
 Number of common numbers in number of form $abc15$ & $15ab5 = 4 \{15015, 15315, 15615, 15915\}$
 $\Rightarrow$ Total number of numbers $= 300 + 30 \times 4 + 34 + 33 - 8 = 487 - 8 = 479$

$$\begin{array}{rcl}
 5) \quad S & = & 1 + 5Z + 9Z^2 + 13Z^3 + \dots + 73Z^{18} \\
 ZS & = & Z + 5Z^2 + 9Z^3 + \dots + 73Z^{19} \\
 - & - & - & - & -
 \end{array}$$

$$S(1 - Z) = 1 + 4(Z + Z^2 + Z^3 + \dots + Z^{18}) - 73Z^{19}$$

$$S(1 - Z) = 1 + 4(-1) - 73 \Rightarrow S = \frac{76}{Z - 1}$$

$$\Rightarrow 19\alpha = 76 \Rightarrow \alpha = 4$$

$$\text{Now } Z = (1)^{1/19}$$

$$S = \frac{76}{((\cos \theta - 1) + i \sin \theta)} \times \frac{((\cos \theta - 1) - i \sin \theta)}{((\cos \theta - 1) - i \sin \theta)}$$

$$\text{where } \theta = \frac{2\pi}{19}$$

$$S = \frac{38[(\cos \theta - 1) - i \sin \theta]}{(1 - \cos \theta)}$$

$$\text{Re}(S) = -38$$

$$\text{Im}(S) = \frac{-38 \sin \theta}{1 - \cos \theta} = -38 \cot \frac{\theta}{2}$$

$$= -38 \cot \left(\frac{\pi}{19} \right)$$

Now compare

$$19\beta = -38 \Rightarrow \beta = -2$$

$$\gamma \cot \left(\frac{\pi}{19} \right) = -38 \cot \left(\frac{\pi}{19} \right) \Rightarrow \gamma = -38$$

6) Option (A)

Let us first of all find the number of digits between 0 and 10^4 having sum of their digits 14.

These numbers have 4 digits say x_1, x_2, x_3, x_4 , then x_1, x_2, x_3, x_4 are integers and $0 \leq x_i \leq 9 \forall i$.

The number of such numbers

$$= \text{Coefficient of } x^{14} \text{ in } (1 + x + x^2 + x^3 + \dots + x^9)^4$$

$$= \text{Coefficient of } x^{14} \text{ in } \left[\frac{1 - x^{10}}{1 - x} \right]^4$$

$$= \text{Coefficient of } x^{14} \text{ in } (1 - x^{10})^4 (1 - x)^{-4}$$

$$= \text{coefficient of } x^{14} \text{ in}$$

$$[1 - {}^4C_1 x^{10} + {}^4C_2 x^{20} - {}^4C_3 x^{30} + {}^4C_4 x^{40}]$$

$$\times (1 + {}^4C_1 x + {}^5C_2 x^2 + {}^6C_3 x^3 + \dots)$$

$$= {}^{17}C_{14} - {}^4C_1 \cdot {}^7C_4 = {}^{17}C_3 - 4 \cdot ({}^7C_3)$$

$$= \frac{17 \times 16 \times 15}{3 \times 2 \times 1} - \frac{4 \times 7 \times 6 \times 5}{3 \times 2 \times 1} = 680 - 140 = 540$$

Also the numbers less than or equal to 10^2 having sum of their digits 14 are 59, 68, 77, 86, 95 which are 5

□ The required number of numbers = $540 - 5 = 535$

Option (B)

Let $\alpha = p^{\alpha_1} q^{\alpha_2} r^{\alpha_3}$

$$\beta = p^{\beta_1} q^{\beta_2} r^{\beta_3}$$

$$\gamma = p^{\gamma_1} q^{\gamma_2} r^{\gamma_3}$$

For p :

Max. of $\{\alpha_1, \beta_1, \gamma_1\}$ is 3

Min. of $\{\alpha_1, \beta_1, \gamma_1\}$ is 1

α_1	β_1	γ_1	
1	3	3	$\longrightarrow 3!/2! = 3$
1	1	3	$\longrightarrow 3!/2! = 3$
1	3	2	$\longrightarrow 3! = 6$

No. of ways to select $(\alpha_1, \beta_1, \gamma_1)$ is 12

for q : min. $\{\alpha_2, \beta_2, \gamma_2\} = 1$

max. $\{\alpha_2, \beta_2, \gamma_2\} = 2$

α_2	β_2	γ_2	
1	2	1	$\longrightarrow 3!/2! = 3$
1	2	2	$\longrightarrow 3!/2! = 3$

No. of ways to select $(\alpha_2, \beta_2, \gamma_2)$ is 6

For r :

α_3	β_3	γ_3	
1	1	1	$\longrightarrow 1 \text{ way}$

No. of ways to select $(\alpha_3, \beta_3, \gamma_3)$ is 1.

$\Rightarrow$ Total ways = $12 \times 6 \times 1 = 72$

Option (C) We know that sum of the numbers formed

by using the digits $d_1, d_2, d_3, \dots, d_n$ is

$$= (n-1)! \left(\sum_{i=1}^n d_i \right) \left[\frac{10^n - 1}{9} \right]$$

□ If $d_1 = 2, d_2 = 4, d_3 = 6, d_4 = 7, d_5 = 9$, then required sum is

$$\left(\frac{(10)^5 - 1}{9} \right) (4!) [2 + 4 + 6 + 7 + 9] = \frac{(10^5 - 1) (4!) (28)}{9}$$

Option (D)

D_n = de arrangement of n persons

$$K = {}^6P_5 (D_6 + D_5) = 6!(265 + 44) = 309 \times 6!$$

$$\begin{aligned} 7) \sum_{r=0}^n \frac{r^2}{nCr} &= a_n + \sum_{r=0}^n \frac{(r-1)(r+1)}{nCr} \\ &= a_n + (n+1) \sum_{r=0}^n \frac{r+2-3}{n+1C_{r+1}} \\ &= a_n + (n+1)(n+2) \sum_{r=0}^n \frac{1}{n+2C_{r+2}} - 3(n+1) \sum_{r=0}^n \frac{1}{n+1C_{r+1}} \end{aligned}$$

$$= a_n + (n+1)(n+2) \left(a_{n+2} - 1 - \frac{1}{n+2} \right) - 3(n+1)(a_{n+1} - 1)$$

$$= a_n + (n+1)(n+2)a_{n+2} - 3(n+1)a_{n+1} - n(n+1)$$

$$8) 1 + \alpha + \alpha^2 + \alpha^3 + \dots + \alpha^{10} = 0$$

$$\bar{\alpha} = \alpha^{10}, \bar{\alpha^2} = \alpha^9, \bar{\alpha^3} = \alpha^8, \bar{\alpha^4} = \alpha^7, \bar{\alpha^5} = \alpha^6$$

$$\text{so } \text{Re}(\lambda + \lambda^2 + \lambda^3 - \lambda^4 + \lambda^5) = \text{Re}(\alpha^6 + \alpha + \alpha^7 + \alpha^2 + \alpha^8)$$

$$= \frac{1}{2} \text{Re}(\alpha^6 + \alpha^5 + \alpha + \alpha^{10} + \alpha^7 + \alpha^4 + \alpha^2 + \alpha^9 + \alpha^8 + \alpha^3) = \frac{1}{2}(-1) = -\frac{1}{2}$$

$$\frac{x^{11} - 1}{x - 1} = (x - \alpha)(x - \alpha^2) \dots (x - \alpha^{10})$$

$$[(x - \alpha^2)(x - \alpha^4)(x - \alpha^6)(x - \alpha^8)(x - \alpha^{10})][(x - \alpha^{12})(x - \alpha^{14})(x - \alpha^{15})(x - \alpha^{18})(x - \alpha^{20})]$$

$$= [(x - \alpha^2)(x - \alpha^4)(x - \alpha^6)(x - \alpha^8)(x - \alpha^{10})][(x - \alpha)(x - \alpha^3)(x - \alpha^5)(x - \alpha^7)(x - \alpha^9)]$$

$$\prod_{k=1}^{10} (x - \beta^k)$$

$$\text{Put } X = \alpha^7 = \mu$$

$$\frac{\alpha^{77} - 1}{\alpha^7 - 1} = \prod_{k=1}^{10} (\mu - \beta^k) = 0$$

$$9) 1 + \alpha + \alpha^2 + \alpha^3 + \dots + \alpha^{10} = 0 \quad \bar{\alpha} = \alpha^{10}, \bar{\alpha^2} = \alpha^9, \bar{\alpha^3} = \alpha^8, \bar{\alpha^4} = \alpha^7, \bar{\alpha^5} = \alpha^6$$

$$\text{so } \text{Re}(\lambda + \lambda^2 + \lambda^3 - \lambda^4 + \lambda^5) = \text{Re}(\alpha^6 + \alpha + \alpha^7 + \alpha^2 + \alpha^8)$$

$$= \frac{1}{2} \text{Re}(\alpha^6 + \alpha^5 + \alpha + \alpha^{10} + \alpha^7 + \alpha^4 + \alpha^2 + \alpha^9 + \alpha^8 + \alpha^3) = \frac{1}{2}(-1) = -\frac{1}{2}$$

$$\frac{x^{11} - 1}{x - 1} = (x - \alpha)(x - \alpha^2) \dots (x - \alpha^{10})$$

$$[(x - \alpha^2)(x - \alpha^4)(x - \alpha^6)(x - \alpha^8)(x - \alpha^{10})][(x - \alpha^{12})(x - \alpha^{14})(x - \alpha^{15})(x - \alpha^{18})(x - \alpha^{20})]$$

$$= [(x - \alpha^2)(x - \alpha^4)(x - \alpha^6)(x - \alpha^8)(x - \alpha^{10})][(x - \alpha)(x - \alpha^3)(x - \alpha^5)(x - \alpha^7)(x - \alpha^9)]$$

$$\prod_{k=1}^{10} (x - \beta^k)$$

$$\text{Put } X = \alpha^7 = \mu$$

$$\frac{\alpha^{77} - 1}{\alpha^7 - 1} = \prod_{k=1}^{10} (\mu - \beta^k) = 0$$

$$10) C_1 : |z - 1| = 1 \Rightarrow (z - 1)(\bar{z} - 1) = 1 \Rightarrow \frac{1}{z - 1} = \bar{z} - 1$$

$$\text{in cartesian form equation of curve } C_1 \text{ is } (x - 1)^2 + y^2 = 1$$

$$\Rightarrow \text{coordinates of any point on } C_1 \text{ are } (1 + \cos\theta, \sin\theta)$$

$$C_2 : w = \frac{z^2 - z - 2}{1 - z} = -z + \frac{2}{z - 1} = -z + 2(\bar{z} - 1)$$

$$\text{Let } z = (1 + \cos\theta, \sin\theta)$$

$$\Rightarrow w = -1 - \cos\theta - i\sin\theta + 2[1 + \cos\theta - i\sin\theta - 1]$$

$$= \cos\theta - 1 - 3i(\sin\theta)$$

$$\Rightarrow \text{Locus of } w \text{ is } (x + 1)^2 + \frac{y^2}{9} = 1$$

$$n - \sqrt{1 - 1} - 2\sqrt{2}$$

$$\text{Let equation of tangent to ellipse be}$$

$$y = m(x+1) + \sqrt{m^2 + 9}$$

□ it is normal to circle

⇒ it passes through (1,0) ⇒ $m^2 = 3$

11) $C_1 : |z-1| = 1 \Rightarrow (z-1)(\bar{z}-1) = 1 \Rightarrow \frac{1}{z-1} = \bar{z}-1$ in cartesian form equation of curve C_1 is $(x-1)^2 + y^2 = 1$

⇒ coordinates of any point on C_1 are $(1 + \cos\theta, \sin\theta)$

$$C_z : w = \frac{z^2 - z - 2}{1 - z} = -z + \frac{2}{z-1} = -z + 2(\bar{z}-1)$$

Let $z = (1 + \cos\theta, \sin\theta)$

$$\Rightarrow w = -1 - \cos\theta - i\sin\theta + 2[1 + \cos\theta - i\sin\theta - 1]$$

$$= \cos\theta - 1 - 3i(\sin\theta)$$

$$\Rightarrow \text{Locus of } w \text{ is } (x+1)^2 + \frac{y^2}{9} = 1$$

$$n - \sqrt{1-1} - 2\sqrt{2}$$

Let equation of tangent to ellipse be

$$y = m(x+1) + \sqrt{m^2 + 9}$$

□ it is normal to circle

⇒ it passes through (1,0) ⇒ $m^2 = 3$

$$\begin{aligned} 12) (1-x^3)^n &= \sum_{r=0}^n a_r x^r \frac{(1-x)^{3n}}{(1-x)^{2r}} \\ \frac{(1-x^3)^n}{(1-x)^{3n}} &= \sum_{r=0}^n a_r \left(\frac{x}{(1-x)^2} \right)^r \\ \frac{(x^2+x+1)^n}{(1-x)^{2n}} &= \sum_{r=0}^n a_r \left(\frac{x}{(1-x)^2} \right)^r \\ \left(1 + \frac{3x}{(1-x)^2} \right)^n &= \sum_{r=0}^n a_r \left(\frac{x}{(1-x)^2} \right)^r \\ \Rightarrow a_r &= 3^r {}^nC_r \end{aligned}$$

$$\sum_{r=0}^{n-1} {}^kC_n 3^r \cdot {}^nC_r$$

Now $r=0$

$$= {}^kC_n (4^n - 3^n)$$

$$\Rightarrow \sum_{n=1}^k ({}^kC_n 4^n - {}^kC_n 3^n)$$

$$= 5^k - 4^k$$

$$\Rightarrow p + q = 9$$

13) $A(k)$ denotes the number of ways in which the man has to climb either 1 step or k steps at a time.

He has to climb $3k$ steps

$$\underbrace{1, 1, 1, \dots, 1}_{3k \text{ times}} = 1 \text{ way}$$

(i) $3k$ times

$$\begin{aligned}
& \underbrace{k, 1, 1, \dots, 1}_{2k \text{ times}} = {}^{2k+1}C_1 \text{ ways} \\
\text{(ii)} \quad & \underbrace{k, k, 1, 1, 1, \dots, 1}_{k \text{ times}} = {}^{k+2}C_2 \text{ ways} \\
\text{(iii)} \quad & \\
\text{(iv)} \quad & k, k, k = 1 \text{ way} \\
\square \text{ Total number of ways} \\
& \Rightarrow 1 + {}^{2k+1}C_1 + {}^{k+2}C_2 + 1 \\
& \because \lambda_k = A(k+1) - A(k) \\
& = 1 + {}^{2k+3}C_1 + {}^{k+3}C_2 + 1 - (1 + {}^{2k+1}C_1 + {}^{k+2}C_2 + 1) \\
& = 2k + 3 + \frac{(k+3)(k+2)}{2} - (2k+1) - \frac{(k+2)(k+1)}{2} \\
& = 2 + \frac{k+2}{2} (k+3-k-1) \\
& \lambda_k = k + 4 \\
& \lambda_{k+1} - \lambda_k = k + 5 - (k + 4) = 1
\end{aligned}$$

$$\begin{aligned}
14) \quad & \square {}^rC_k {}^{2024}C_r = \frac{2024!}{(r-k)!k!(2024-r)!} \\
& \sum_{k=0}^{1012} {}^{1012}C_k \sum_{r=k}^{2024} \frac{(2024-k)!}{(r-k)!(2024-r)!} \\
\text{Now} \quad & \sum_{k=0}^{1012} {}^{1012}C_k \sum_{r=k}^{2024} {}^{2024-k}C_{r-k} \\
& = \sum_{k=0}^{1012} {}^{1012}C_k 2^{2024-k} \\
& = {}^{1012}C_0 2^{2024} + {}^{1012}C_1 2^{2023} + \dots + {}^{1012}C_{1012} 2^{1012} \\
& = 2^{1012} [{}^{1012}C_0 2^{1012} + {}^{1012}C_1 2^{1011} + \dots + {}^{1012}C_{1012} 2^0] \\
& = 2^{1012} (2+1)^{1012} = 6^{1012} \\
\text{Now } 6^{1012} & = (7-1)^{1012} \\
& \text{When divided by 49 remainder is 22.}
\end{aligned}$$

$$\begin{aligned}
15) \quad & I_m (z_1^2 + z_2^2 + z_3^2) = 2(x_1y_1 + x_2y_2 + x_3y_3) \quad \dots(1) \\
& |z_1 z_2 z_3| = |z_1| |z_2| |z_3| = \sqrt{8} \\
& \Rightarrow |z|^3 = \sqrt{8} \quad (|z_1| = |z_2| = |z_3|) \\
& \Rightarrow |z| = \sqrt{2} \\
& z_1 + z_2 + z_3 = -\frac{\sqrt{3}}{2} - \sqrt{5}i \\
& \text{conjugate both sides} \\
& \bar{z}_1 + \bar{z}_2 + \bar{z}_3 = -\frac{\sqrt{3}}{2} + \sqrt{5}i \\
& \Rightarrow \frac{2}{z_1} + \frac{2}{z_2} + \frac{2}{z_3} = -\frac{\sqrt{3}}{2} + \sqrt{5}i \\
& \Rightarrow \sum z_1 z_2 = -\frac{15}{4} + \frac{\sqrt{15}}{4}i \\
& \text{Now } z_1^2 + z_2^2 + z_3^2 = (z_1 + z_2 + z_3)^2 - 2\sum z_1 z_2 \\
& = \frac{13}{4} + \frac{\sqrt{15}}{2}i
\end{aligned}$$

$$\Rightarrow I_m(z_1^2 + z_2^2 + z_3^2) = \frac{\sqrt{15}}{2}$$

put in equation (1)

$$\Rightarrow x_1 y_1 + x_2 y_2 + x_3 y_3 = \frac{\sqrt{15}}{4}$$

16) In A, general term of the series is

$$\begin{aligned} T(r) &= \frac{{}^{50+r}C_r (2r-1)}{{}^{50}C_r (50+r)} \\ &= \frac{{}^{50+r}C_r}{50C_r} \left(1 - \frac{50-r+1}{50+r} \right) \\ &= \frac{{}^{50+r}C_r}{50C_r} - \frac{{}^{50+r}C_r}{50C_r} \left(\frac{50-r+1}{50+r} \right) \\ &\text{Now, } \frac{{}^{50+r}C_r}{50C_r} \left(\frac{50-r+1}{50+r} \right) \\ &= \frac{(50-r+1)(50+r)!r!(50-r)!}{r!50!(50+r)50!} \\ &= \frac{(50-r+1)!(50+r-1)!}{50!50!} \\ &= \frac{{}^{50+r-1}C_{r-1}}{{}^{50}C_{r-1}} \\ \Rightarrow T(r) &= \frac{{}^{50+r}C_r}{50C_r} - \frac{{}^{50+r-1}C_{r-1}}{{}^{50}C_{r-1}} = V(r) - V(r-1) \end{aligned}$$

where $V(r) = \frac{{}^{50+r}C_r}{50C_r}$

Now, sum of the given series

$$\begin{aligned} A &= \sum_{r=1}^{50} T(r) = V(50) - V(0) \\ &= \frac{{}^{100}C_{50}}{{}^{50}C_{50}} - \frac{{}^{50}C_0}{{}^{50}C_0} = {}^{100}C_{50} - 1 \\ &= \sum_{r=0}^{50} \left({}^{50}C_r \right)^2 = {}^{50}C_0^2 + {}^{50}C_1^2 + {}^{50}C_2^2 + \dots + {}^{50}C_{50}^2 \end{aligned}$$

Also, $B = \sum_{r=0}^{50} {}^{50}C_r$

$$\begin{aligned} C &= \sum_{r=1}^{49} \frac{2r^2 - 48r + 1}{(50-r) \cdot {}^{50}C_r} = 50 \sum_{r=1}^{49} \frac{(r+1)^2 - r(50-r)}{(50-r) \cdot {}^{50}C_r} \\ &= 50 \sum_{r=1}^{49} \left(\frac{(r+1)^2}{(50-r) \cdot {}^{50}C_r} - \frac{r}{50C_r} \right) \\ &= 50 \sum_{r=1}^{49} \left(\frac{r+1}{(50-r) \left(\frac{{}^{50}C_r}{r+1} \right)} - \frac{r}{50C_r} \right) \\ &= 50 \sum_{r=1}^{49} \left(\frac{r+1}{50C_{r+1}} - \frac{r}{50C_r} \right) \\ &= 50 \left(\frac{50}{50C_{50}} - \frac{1}{50C_1} \right) = 2499 \end{aligned}$$

$$\Rightarrow A - B + C = 2498$$

$$17) \frac{(x^{18} - 1)^2}{(x - 1)^2} - x^{17} = \frac{(x^{36} - 2x^{18} + 1) - (x^{19} - 2x^{18} + x^{17})}{(x - 1)^2}$$

$$\frac{x^{36} - x^{19} + 1 - x^{17}}{(x - 1)^2} = \frac{(x^{17} - 1)(x^{19} - 1)}{(x - 1)^2}$$

$$P(x) = 0 \Rightarrow x = e^{i\frac{2\pi k_1}{17}}, e^{i\frac{2\pi k_2}{19}}$$

$$k_1 = 1, 2, \dots, 16 \text{ and } k_2 = 1, 2, \dots, 18$$

$$\sum_{r=1}^5 a_r = \frac{1}{19} + \frac{1}{17} + \frac{2}{19} + \frac{2}{17} + \frac{3}{19} = \frac{159}{323}$$

$$\Rightarrow m + n = 482$$

PART-2 : PHYSICS

18)

Correct Answer is (C)

19)

Correct Answer is (C)

20)

Correct Answer is (A)

21)

Correct Answer is (A)

22)

Correct Answer is (B,C)

23)

Correct Answer is (A,B,C)

24)

Correct Answer is (A,B,C,D)

25)

Correct Answer is (0.18 to 0.20)

26)

Correct Answer is (1.60)

27)

Correct Answer is (7.87)

28)

Correct Answer is (0.48)

29)

Correct Answer is (14)

30)

Correct Answer is (203)

31)

Correct Answer is (8)

32)

Correct Answer is (6)

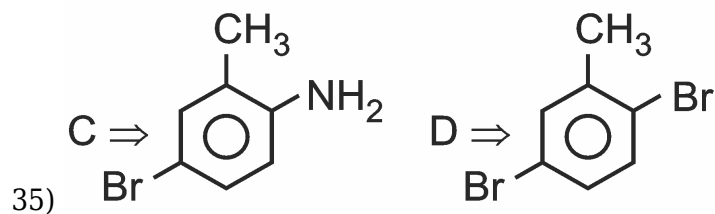
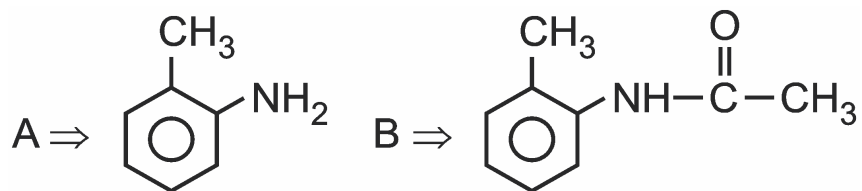
33)

Correct Answer is (892)

34)

Correct Answer is (3)

PART-3 : CHEMISTRY



Compound (A) can be ortho/meta toluidine
 Option (C) - Gatterman reaction
 Option (D) - Ortho effect (SIP)

36)

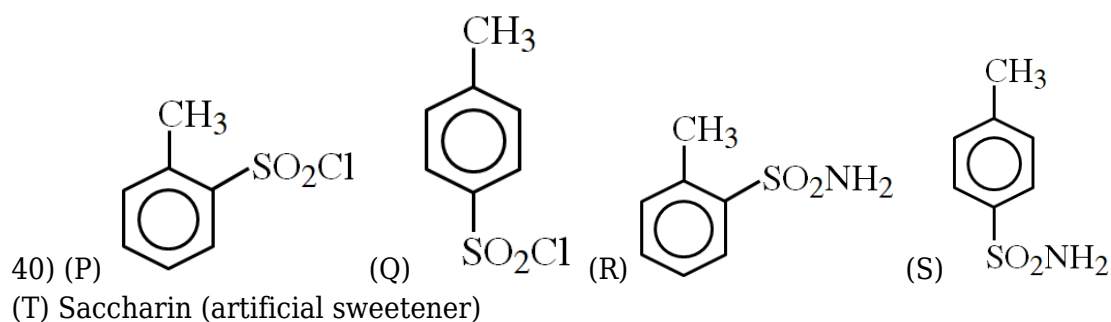
The correct answer is option (A)

37) (A) Acetanilide
 (C) Aryl halide do not give SN_2
 (D) Acidic medium

38) (A) both +ve
 (B) both -ve
 (C) both -ve
 (D) Alanine -ve, Insulin +ve

39)

The correct answer is (A), (C), (D)



41)

(B) HDP is formed instead of LDP.

42) Terpeneol (2 + 3)

43) Chloroxylenol (3 + 4 + 5)

44) P = Salicylic acid, Q = Acetic acid, R = Acetyl chloride, S = Propanoic acid, T = EtBr, U = EtNC, V = HCOOH, W = EtNH₃⁺, X = EtNH₂, Y = EtOH

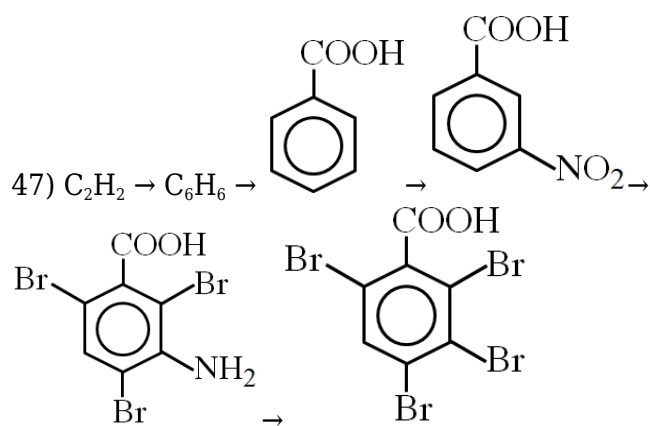
45) A = CH₃NHCOPh

46) P is o-nitrophenol

Q is p-nitrophenol

R is Ag

S is o-nitrosophenol (Mulliken Barker test for -NO₂)



$\frac{1}{3}$ moles (Y) are produced $\square$ mass = $(1/3) \times 438 = 146$ g.

48) (CH₃)₂C(Br)COOH

49)

(1), (5), (6), (8), (9)

50) 6-oxygen, 2-nitrogen

51) Ala, Ser, Asp, Cys (all are non essential)